Suxamethonium

Mechanism of action

- Suxamethonium is a neuromuscular blocker. It antagonises (blocks) nicotinic acetylcholine receptors at the neuromuscular junction (motor nerve end plate) of skeletal muscle. This results in the inability of skeletal muscles to contract.

Authorised practice levels

- CCPs.

Indications

- ✓ Neuromuscular blockade for rapid sequence intubation (RSI).

Contraindications

- ✕ Known severe allergy.
- ✕ Known personal or family history of malignant hyperthermia (MH). In susceptible patients suxamethonium will trigger life-threatening MH.
- ✕ Pre-existing paraplegia or quadriplegia. Long-term muscle weakness causes proliferation of acetylcholine receptors on skeletal muscle and suxamethonium may cause life-threatening hyperkalaemia.
- ✕ Muscle disorder with long-term weakness. Examples include muscular dystrophy and motor neurone disease. Long-term muscle weakness causes proliferation of acetylcholine receptors on skeletal muscle and suxamethonium may cause life-threatening hyperkalaemia.
- ✕ Hyperkalaemia is strongly suspected. Suxamethonium will cause a brief rise in the potassium concentration within blood of 0.5-1 mmol/litre. In the presence of hyperkalemia this rise may be significant.

Cautions

- ⚠ Predicted difficult intubation.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

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Dosage

- See the 'RSI' guideline.
- The onset of adequate neuromuscular blockade may be delayed if cardiac output is very low. Consider increasing the dose of suxamethonium to approximately 3 mg/kg, up to a maximum of 200 mg.

Administration

- Administer undiluted for an adult or a large child.
- Dilute 100 mg to a total volume of 10 ml for a small child. This solution contains 10 mg/ml.
- Administer IV as a bolus, preferably into a running IV line.

Common adverse effects

- Bradycardia. This is due to stimulation of acetylcholine receptors in the parasympathetic nervous system and is usually only clinically significant in young children.

Usual onset of effect

- 30-60 seconds. This is predominantly affected by cardiac output and will be prolonged if cardiac output is low.

Usual duration of effect

- 4-8 minutes.

Usual preparation

- Ampoule containing 100 mg in 2 ml.

Pharmacokinetics

- Suxamethonium is metabolised by the enzyme pseudocholinesterase. In most patients suxamethonium is cleared within 4-8 minutes.

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- The neuromuscular blocking effects of suxamethonium can be increased to 4-12 hours if the patient has pseudocholinesterase deficiency.
- Pseudocholinesterase deficiency is rare.

Common interactions

- None.

Additional information

- Suxamethonium that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is 8 weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Suxamethonium is a depolarising neuromuscular blocker. This means that the acetylcholine receptor at the neuromuscular junction is stimulated prior to being blocked and muscle fasciculations occur prior to the onset of neuromuscular blockade.
- Malignant hyperthermia (MH) is a rare, inherited disorder of muscle metabolism affecting approximately 20 families in New Zealand, many of whom are in the Manawatu area. When exposed to some anaesthetic agents (including suxamethonium) the patient may develop a life-threatening hypermetabolic state with severe hyperthermia. Patients with MH (or a family history of MH) usually know about it.
- Burn injury is often quoted as a contraindication to the administration of suxamethonium because life-threatening hyperkalaemia may occur. However, this is not a contraindication immediately following burn injury or once the patient is well enough to be discharged from hospital.

More Information

Version history

22 November 2022 1.0.1

Update: Discard date following removal from the fridge.